

PRISM-4D GLP-1R / Aleniglipron Pharmacological Dynamics Intelligence Report

Scientific / program-lead report on M2 dynamics evidence

Confidential Discussion Material

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Use of this document. This package presents computational translational evidence from the PRISM-4D engine and downstream analysis layers. It is *not* a clinical-effect claim, *not* a patient-response prediction, and *not* experimental validation. Assay recommendations are falsification gates, not confirmation ceremonies. PROJECTED and HYPOTHESIZED items require wet-lab falsification before any biological assertion.

Epistemic legend. Every claim, table row, and figure inherits exactly one of the following classes, which governs how it may be cited:

- **OBSERVED** direct tensor measurement from PRISM-4D engine outputs.
- **DERIVED** deterministic transform of observed tensors.
- **INFERRED** multi-tensor interpretation; not a single-tensor measurement.
- **PROJECTED** translational extrapolation beyond simulated conditions.
- **HYPOTHESIZED** requires wet-lab falsification before any biological claim.

1 M2 Pharmacological Dynamics Intelligence Report

1.1 1. Executive Summary & Epistemic Boundaries

This dossier summarizes an 80-replica, 1,600-stream receptor-side analysis of the GLP-1R Aleniglipron program, including the 5.62-billion spike extraction and downstream tensor fusion.

Strict Boundary: This dossier provides computationally bounded receptor-side mechanistic evidence. It does not claim in vivo clinical durability. All findings are subject to the explicit wet-lab falsification gates defined in Section 7.

Primary outputs:

Output	Value
Calibration anchors screened	512
Zero-shot replacements selected	10
Liability edge	glp1r_5VEX_WT:PHE143->TYR148
Steric-clean gate	$E[\text{Pi_clash}] \leq 0.10$
SA gate	$\text{SA} \leq 3.5$

1.2 2. Thermodynamic Manifold & Multi-Modal Convergence

Phase-Manifold Coherence was computed from the 5-phase thermal protocol and stream-level phase counts. Shannon entropy and concordance across orthogonal perturbation streams are used to distinguish constitutive receptor features from thermal artifacts.

Metric	Value
Mean residue coherence score	0.3779
Median Shannon entropy channel	0.9656
Maximum active streams observed	20
Maximum active phases observed	5

Edge	Class	Status	Edge coherence	From entropy	To entropy	From concordant channels	To concordant channels
PHE143 -> ILE147	pocket_vect or	validated_co nstitutive	0.5195	0.9666	0.9661	5/7	4/7
PHE169 -> SER416	downstream _lock	validated_co nstitutive	0.5090	0.9654	0.9651	5/7	5/7
TYR241 -> ARG310	downstream _lock	validated_co nstitutive	0.5090	0.9653	0.9651	4/7	5/7
ARG421 -> TRP417	downstream _lock	validated_co nstitutive	0.4894	0.9656	0.9654	4/7	4/7
TRP417 -> ARG421	downstream _lock	validated_co nstitutive	0.4894	0.9654	0.9656	4/7	4/7

1.3 3. Transition Chronology & Allosteric Wiring

The translation-pathway tensor identifies ASN182 as a load-bearing intermediate in the receptor wire. Mechanical load is represented as the force-aligned scalar $F \cdot ASC$; pathway rows below are sorted by wire score and load support.

Condition	Rank	Residue	Mean absolute load	Shear stress	Structural high-gradient flag	High kinetic burst flag	Wire score
glp1r_5VEX_WT	1	ASN182	2381.9366	4.0178	True	False	0.3439
glp1r_6X1A_WT	1	ASN182	1726.8185	4.4062	True	False	0.3391
glp1r_6XOX_WT	1	ASN182	1824.2020	5.3400	True	False	0.2822
glp1r_6XOX_A316T	1	ASN182	1899.5024	5.0121	True	False	0.2771
glp1r_6X1A_WT	2	PHE367	1506.7656	5.2064	True	False	0.2429

The temporal cascade table records first ramp firing order. The transition chronology is reported as a predicted early-transition delay relative to initiation-wave residues; the table reports the observed `first_ramp_md_step` values used to anchor the ordering without asserting absolute biological timing.

Condition	Protocol	Residue index	First ramp MD step	Ramp-up spikes	Supporting streams
glp1r_5VEX_W T	A_ThermalShock	150	10021.410	24252	6
glp1r_5VEX_W T	A_ThermalShock	421	10021.410	44391	6
glp1r_5VEX_W T	A_ThermalShock	182	10026.406	79682	6
glp1r_5VEX_W T	A_ThermalShock	417	10031.402	58780	6
glp1r_5VEX_W T	A_ThermalShock	7	10036.397	248118	6
glp1r_5VEX_W T	A_ThermalShock	46	10036.397	109763	6
glp1r_5VEX_W T	A_ThermalShock	12	10041.393	242702	6
glp1r_5VEX_W T	B_UVAromatic	182	8000.999	51974	4
glp1r_5VEX_W T	B_UVAromatic	7	8000.999	157445	4
glp1r_5VEX_W T	B_UVAromatic	421	8000.999	28809	4
glp1r_5VEX_W T	B_UVAromatic	46	8000.999	68781	4
glp1r_5VEX_W T	B_UVAromatic	150	8000.999	15609	4
glp1r_5VEX_W T	B_UVAromatic	417	8000.999	37900	4
glp1r_5VEX_W T	B_UVAromatic	12	8000.999	152125	4
glp1r_5VEX_W T	C_Equilibrium	421	6000.571	100495	6
glp1r_5VEX_W T	C_Equilibrium	182	6000.615	181591	6
glp1r_5VEX_W T	C_Equilibrium	12	6012.911	548412	6
glp1r_5VEX_W T	C_Equilibrium	417	6019.059	133494	6
glp1r_5VEX_W T	C_Equilibrium	46	6023.409	245695	6
glp1r_5VEX_W T	C_Equilibrium	150	6185.060	55133	6
glp1r_5VEX_W T	C_Equilibrium	7	6197.356	562442	6
glp1r_5VEX_W T	D_Hysteresis	150	6095.347	23938	4
glp1r_5VEX_W T	D_Hysteresis	421	6095.347	44787	4
glp1r_5VEX_W T	D_Hysteresis	182	6100.200	79162	4

Mechanical load summary:

Condition	Observations	Mean mechanical load	P99 mechanical load
glp1r_5VEX_WT	1396000	1830.6665	13621.2217
glp1r_6LN2_A316T	1419800	1708.5544	13493.4326